Preparing for the Supervisor Call

Interneuron-Differentiated Inhibition in a Laminar SDR Column — thesis proposal discussion

Subham Singh (1506413) · Frankfurt UAS · Prep briefing

The single most important thing. He is not testing whether you already know everything. He is testing whether you understand what you proposed and can think on your feet about it. "I haven't worked that out yet, but here's how I'd approach it" is a strong answer for a master's student. Pretending to know something you don't is the only real way to lose credibility — he wrote this framework and will catch it instantly. A long call is a good sign: it means he's taking the project seriously.

This covers six blocks: (1) the pitch, (2) the conceptual core, (3) the honest-novelty story, (4) his likely challenges, (5) your open decisions, (6) repo readiness. Skim the blue "say it like this" boxes the morning of the call — they're the lines that should come out automatically.

BLOCK 1 The Pitch

What he tests in the first two minutes. State it cleanly and the call becomes a discussion; fumble it and he probes whether you understand your own project.

The two-sentence version

> SAY IT LIKE THIS
>
> *HTM and the whole SDR tradition collapse all cortical inhibition into a single k-winner-take-all rule that just fixes sparsity. My thesis replaces that with two mathematically distinct interneuron operators — a divisive-normalisation PV and a slow dendritic-gating SST — inside one laminar column, and tests whether they produce a functional split between encoding and retrieval that kWTA can't.*

The one-paragraph expansion (if he wants more)

> SAY IT LIKE THIS
>
> *The biology of PV and SST doing different jobs is well established, and so is each mechanism on its own — divisive normalisation for PV, dendritic gating for SST. What nobody has done is bring those two distinct operators into the discrete-SDR/HTM substrate and test an encoding-versus-retrieval dissociation there. So it's a synthesis contribution, not a new mechanism — and the first thing I do is a falsification check to prove the PV operator is genuinely different from kWTA before I build anything on top of it.*

How to deliver it

- **Lead with the gap, not the mechanism.** "HTM only has one inhibition rule" is the hook that makes him nod. Don't open with "I'm modelling parvalbumin as divisive normalisation" — that's detail before he knows why it matters.
- **Say the terms naturally, not carefully.** Use "kWTA" and "divisive normalisation" like you own them. He knows them cold; you don't need to define them.

- **Have "dissociation" ready** and know it means: PV mainly affects encoding, SST mainly affects retrieval — *differential, not exclusive*. That qualifier signals you understand it's a nuanced claim, not a crude double-dissociation.
- **Don't oversell.** The pitch above is built to be un-correctable — every claim survives scrutiny because it's honestly scoped. Practise the two-sentence version out loud until it comes out the same way each time.

BLOCK 2 The Conceptual Core

The four ideas he'll expect you to explain in your own words, not recite. He'll push past the *what* into the *why*. Everything below traces back to one idea: **PV and SST are different in kind, and that difference is what buys the dissociation.**

1. What kWTA is, and why it's a limitation

> **SAY IT LIKE THIS**
>
> *k-winner-take-all just picks the top-k most-active units and switches everything else off, so sparsity is fixed by hand — you decide k in advance. It's one knob. It doesn't matter how strong or weak the input is; you always get the same number of winners.*

> **THE WHY HE'S AFTER**
>
> The limitation isn't that kWTA is wrong — it's that it's homogeneous and rigid. Real cortex has no fixed k; assembly size responds to how much drive there is. And one rule can't stand in for what several interneuron types do. That's the opening the whole thesis walks through.

2. The PV operator (divisive normalisation) and how it differs from kWTA

> **SAY IT LIKE THIS**
>
> *PV I model as divisive normalisation: each unit's activity gets divided by the total drive across the population, then thresholded. So instead of ranking and cutting at k, the whole population's gain is scaled by how much total input there is.*

> **THE CRITICAL DIFFERENCE — HE WILL ASK THIS**
>
> kWTA can never produce an empty assembly. Divisive normalisation can. If total drive is weak or ambiguous, everything can fall below threshold and you get zero winners — kWTA structurally always returns k. And divisive normalisation lets assembly size *vary with drive*, which kWTA's fixed k forbids. Those two behaviours — variable size and reachable full suppression — are exactly what prove the operator does something kWTA can't.

> **SUBTLETY TO HAVE READY**
>
> In the pure divisive form the denominator is the same for every unit, so λ alone doesn't change the *ranking* — it doesn't decide winners. What makes it differ from kWTA is the **fixed threshold combined with the normalisation**: that's what lets the population go silent and lets size track drive. If he asks "isn't this just a smooth kWTA?", that interaction is your answer — and the falsification check is how you prove it empirically rather than assert it.

3. The SST operator (slow dendritic gating) and why it's different in kind

SAY IT LIKE THIS

SST I model as a slow leaky integrator that gates the apical stream — the top-down / retrieval input — before it's combined with the feedforward drive. The key thing is it has memory: it integrates over time, so it changes slowly.

THE WHY

PV is stateless and fast — it responds to the current step's drive and nothing else. SST is stateful and slow — it carries a running average, so it stabilises across time. That difference in *kind* (memoryless vs memory, fast vs slow, somatic vs dendritic) is what makes them two genuinely distinct operators, not two copies of one idea with different labels. It's also *why* they split along encoding vs retrieval: fast per-step gain control suits encoding sparsity; slow persistent gating suits keeping a retrieved pattern stable.

4. Why the falsification check exists — and why it's first

SAY IT LIKE THIS

The whole thesis rests on PV being functionally different from kWTA. If my PV parameters happen to sit in a regime where it behaves just like kWTA, I've relabelled the old mechanism and proven nothing. So before I build anything on top, I run a check: does assembly size vary with drive? Is full suppression reachable? Is the graded pre-threshold value actually used? If those fail, I stop and revisit before committing to the rest.

Why he'll love this. It means you've thought about how your own project could produce a *fake positive*, and built a gate to catch it. That's the difference between a student who wants the result and a researcher who wants the truth. If you land nothing else, land this.

ON PARAMETERS (e.g. if he asks "what λ will you use?")

The correct answer is: "That's one of the parameters I'll tune during Sim 1 — and the falsification check is partly how I'll know I've picked a regime that's actually doing something." λ, θ, ε are listed as "open parameters to fix during implementation" in his own issue — so this isn't a gap in your knowledge, it's you knowing the plan.

BLOCK 3 The Honest-Novelty Story

The block students most often mishandle, because talking about what's *not* novel feels like admitting weakness. It's the opposite: to someone who wrote the framework and knows the literature, honest scoping is the strongest signal of competence. Overclaiming gets caught instantly.

The move: "here's what's established, here's the one unclaimed combination"

SAY IT LIKE THIS

I'm not claiming a new biological mechanism. The PV/SST division of labour, divisive normalisation for PV, SST dendritic gating, laminar PV/SST columns — all of that exists. What's unclaimed is the specific

> *combination: bringing two mathematically distinct interneuron operators into the HTM/SDR substrate, which has only ever used homogeneous kWTA, and testing an encoding-versus-retrieval dissociation over discrete SDRs. It's a synthesis / substrate-transfer contribution, and I'd rather state that honestly than overclaim.*

Know exactly who has done what — so you're never caught flat

- **Hertäg & Sprekeler (2019/2020)** — already dissociate PV (somatic) vs SST (dendritic) computationally. *But* in a rate model, for prediction-error formation, not over SDRs and not on an encoding/retrieval axis. This is the paper most likely to be thrown at you as "hasn't this been done?"
- **Krishnamurthy, Silberberg & Lansner (2012)** — already dissociate basket (PV-like) vs Martinotti (SST-like) in an attractor column. *But* their axis is onset-versus-termination (dwell time), not encoding-versus-retrieval, and their PV is depressing-synapse basket cells, not a divisive-normalisation operator.
- **Moreni / Mejias (2025)** and recent predictive-coding column models — laminar columns with PV/SST/VIP. *But* spiking / biophysical, no SDRs, and the question is feedforward/feedback firing, not encoding/retrieval.
- **Fransén & Lansner (1998)** — a classical associative-memory column with storage and retrieval. *But* homogeneous inhibition; no operator-distinct interneurons.

The one-line defence to memorise. "Yes — the dissociation exists in rate and attractor models. What doesn't exist is it in a discrete-SDR substrate, on an encoding-versus-retrieval axis, with a falsification test that the operator isn't just kWTA in disguise. That's the gap." Saying this *before* he has to corner you is what turns a weakness into a display of command.

BLOCK 4 His Likely Challenges & How to Answer

The specific hard questions he's positioned to ask — from the science and from his own repo. For each, the honest, credible answer.

> **"Isn't divisive normalisation just a smooth version of kWTA?"**
>
> No — and that's exactly what the falsification check proves. With a fixed threshold, divisive normalisation can drive the whole population silent (empty assembly) and lets assembly size vary with total drive. kWTA can do neither; it always returns k. If my parameters ever sit in a regime where those behaviours vanish, I report that I haven't yet shown a functional difference — that's the go/no-go gate.

> **"Haven't Hertäg & Sprekeler already dissociated PV and SST?"**
>
> Yes, in a rate model, for prediction-error neurons. What's different here is the substrate and the question: discrete SDRs rather than continuous rates, and an encoding-versus-retrieval axis rather than prediction error. I cite them explicitly and position against them — the contribution is the transfer, not the dissociation itself.

“Why rate-coded and not spiking? Doesn’t that weaken biological plausibility?”

It does trade away some biophysical realism, and I’m explicit that spiking / Hodgkin-Huxley detail is out of scope. The reason is that HALO is SDR-native and the question I’m asking — do two distinct inhibitory operators dissociate over SDR assemblies — lives at the algorithmic level, not the biophysical one. Spiking would add cost without changing the answer to that specific question. It’s a deliberate scoping choice, not an oversight.

“How do you measure timing precision (H1b) when the substrate has no time within a step?”

That’s exactly why H1b is conditional. As written, the model has no within-step temporal dimension, so I’d add a coarse one — activation-order or phase — to make timing measurable. If that extension proves infeasible in the timeframe, H1 falls back to assembly-size control alone (H1a) and timing becomes future work. I kept the fallback precisely so the claim stays falsifiable rather than untestable.

“What’s your baseline? How do you know a dissociation is real and not noise?”

The plain HTM column (existing kWTA CorticalUnit) is the reference baseline throughout. For H4 the test is a graded PV×SST sweep, so I’m looking at effect *curves*, not a single on/off point: PV should dominate the encoding metric and SST the retrieval metric, with neither being exactly zero on the other’s — differential, not exclusive. That pattern across the sweep is much harder to produce by noise than a single contrast.

“What if the falsification check fails — the PV operator behaves like kWTA?”

Then I stop and revise the conceptual model before touching Sim 2 or Sim 3, per the decision rule. It would mean this iteration hasn’t demonstrated a functional difference yet — which is still a legitimate, writable finding. A thesis documents what I found; a well-run negative result is a result, not a failure.

“How does a single column connect to my broader heterarchical / multi-column vision?”

The single laminar column is the building block. Cross-column plasticity and multi-column voting are explicitly out of scope for this thesis, but this is exactly the substrate they’d build on — a column with cell-type-differentiated inhibition is a richer unit for the heterarchical layer to coordinate than a plain kWTA column. So it feeds the larger programme without over-committing this thesis to it.

“Where does this plug into HALO, concretely?”

As a new `LaminarCorticalUnit` extending `CorticalUnitBase`, config-switchable against the existing plain `CorticalUnit` — following the framework’s ‘swappable via config, not hard imports’ convention. L4 is a graded relay (no sparsification), L2/3 is where PV and SST act, L5 supplies the retrieval cue, L6 is a gain-control placeholder. The synthetic streams come from the contextual-task suite already on the phase-2 branch.

BLOCK 5 Your Open Decisions — Come With Recommendations

You handed him three scope calls. Don't just receive verdicts — walk in with a recommendation for each, so you're steering the conversation. He may overrule you; that's fine. Having a view is what matters.

Decision 1 — H1b (timing)

> **YOUR RECOMMENDATION**
>
> Keep it as committed work *with* the assembly-size-only fallback. Adding a coarse temporal dimension is worth it because timing is half of what PV is known for — but don't front-load it. Build and validate the core (H1a assembly size) first; add the temporal dimension in WP4 once the falsification gate has passed. If he thinks a temporal dimension is more than a master's should take on, be ready to drop it to future work without argument.

Decision 2 — Sim 3 / H3 (arbitration)

> **YOUR RECOMMENDATION**
>
> Your instinct promoted all three sims to committed. Defensible, but say clearly that **Sim 1 + Sim 2 are the guaranteed core** and Sim 3 is the highest-value, highest-risk piece — the most likely basis for a paper. Offer him the choice: keep Sim 3 committed, or hold it as a stretch with a clean fallback. Framing it this way shows you understand the risk profile, which is more impressive than insisting on all three.

Decision 3 — datasets & thresholds

> **YOUR RECOMMENDATION**
>
> Propose NAB (NYC Taxi + one realKnownCause stream) as the HTM-native benchmark and UCI HAR as the multi-context stream, both provisional. Thresholds: ~2% ± 0.5% encoding sparsity (and drive-dependent), ≥80% retrieval SDR overlap, ≤5% degradation over ≥10 cycles, robustness to ~10–20% noise. Present these as starting points and ask if he'd set them differently — he may have specific numbers in mind from his own runs.

> **NOTE**
>
> The acceptance thresholds are not currently in the proposal document (they were dropped when the Data section moved). So if he opens the proposal looking for them, they're not there — raise them verbally, and offer to add a one-line thresholds sentence to the Data section afterwards.

BLOCK 6 Repo Readiness

The parts of HALO you should be fluent in, because they're where your work plugs in. He may open any of these. You don't need to have memorised the code — you need to be able to talk about what each piece does and how yours attaches.

The existing CorticalUnit — what you're extending

- Today's `CorticalUnit` is one undifferentiated population: a potential-pool / permanence pattern feeding a single `_winner_take_all()` step that fixes sparsity. No layers, no PV, no SST. Your `LaminarCorticalUnit` replaces that one step with the laminar structure and the two operators.
- Be able to name the four sub-pools from the Sim 1 issue: `L4Pool` (graded relay, reuses potential-pool/permanence, *no* WTA), `PVPool` (stateless divisive normalisation + threshold), `SSTPool` (stateful leaky integrator gating the apical stream), `L23Pool` (combines basal + SST-gated apical, then PV replaces WTA).

Why the L4/L2-3 split matters (a likely probe)

- L4's output must stay *graded*, not sparse. If L4 sparsified first, PV would have no competitive landscape to act on and the whole thing collapses back into today's single-step `encode()` with extra labels. Knowing this shows you understand *why* the architecture is shaped the way it is, not just what it contains.

The "swappable via config, not hard imports" convention

- New components are selected by config, never hard-imported, so `LaminarCorticalUnit` can be switched back to the plain `CorticalUnit` for baseline comparison. This is a repo-wide convention he cares about — mention you'll follow it.

The contextual-task suite (phase-2 branch) — your synthetic data source

- A pluggable `Task` abstract base + config + decorator registry, with three tasks forming a ladder: `RandomNoiseTask` (negative control), `SequenceTask` (single-context cyclic symbols), `ContextSwitchTask` (interleaved multi-context, where context must be inferred, not read off the symbol). These generate your controlled synthetic streams.

Where the column sits in the bigger HALO picture (background.md)

- Populations of `CorticalUnit`s are coordinated by a `HeterarchicalLayer` (learned lateral biases that modulate, not drive), a `ThalamicLayer` (reliability-weighted aggregation), a `TRNGatingLayer` (entropy-based gating), a `ReliabilityModule` (trust), and a `ConsensusEngine` (weighted-vote consensus). You don't touch these — but knowing your column is the unit they orchestrate lets you answer the "how does this fit the vision" question with precision.
- One honest note if it comes up: the `TRNGatingLayer` is also where Sim 3's apical-arbitration mode would live (per the Sim 3 issue), operating *inside* the column, keeping the existing network-level gating as-is.

> **If he opens a file you don't know well:** say so plainly and reason from principles — "I haven't gone deep into that module yet; my understanding is it does X, and my work would attach at Y." That's credible. Bluffing about his own code is the one thing guaranteed to go badly.

Final checklist for the morning of the call

- Two-sentence pitch — automatic, same every time.
- "Empty assembly + size varies with drive" — the two things kWTA can't do.

- “Differential, not exclusive” — the dissociation qualifier.
- The falsification check — why it’s first, what it’s guarding against.
- Hertäg & Sprekeler + Krishnamurthy — who did the dissociation, and how yours differs.
- A recommendation ready for each of the three open decisions.
- Thresholds aren’t in the proposal — raise them verbally.